Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Plane shapes: types, properties and perimeter. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	2
Topic	Plane Shapes (2-D Shapes)
Subtopic	Plane shapes: types, properties and perimeter
Theme	—
Curriculum objectives	• identify the types of plane shapes and state their properties, similarities and differences • identify the parts of a circle • find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle
Previous knowledge	• Students have been taught angles.
Lesson sequence	• identify the types of plane shapes and state their properties, similarities and differences — the teacher draws the various shapes on the board; the student identify their properties; the teacher writes the properties of the shapes on the board for the students to learn • identify the parts of a circle — the teacher explains the parts of a circle to the students: the arc, radius, chord and diameter, with the circumference, segment, sector and tangent labelled on the board drawings • find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle — the teacher works the perimeter of plane shapes on the board: square, rectangle, trapezium, parallelogram, regular hexagon and circle, including finding a missing side or radius when the perimeter is given
Instructional materials	• Cardboard containing angles presentation.
Evaluation	• Find the perimeter of a square of sides 8cm • Find the perimeter of a rectangle of length 7cm and breadth 4cm. • What is the perimeter of regular pentagon with sides 5cm. • How many triangles can an octagon be divided into if you draw diagonals from one vertex to all non-adjacent vertices? • Find the perimeter of a parallelogram whose base is 6cm and sides 5cm • The circumference of a circle is 22cm – Find the radius of a circle ($\pi = \frac{22}{7}$).

	• A rectangle is such that its length is twice its breadth, if the breadth is d, find the total length of the sides of rectangle. • Calculate the perimeter of the figure below. [the figure is not included here, marked 1cm, 5cm, 9cm] • Find the perimeter of a regular hexagon with sides 5cm.

Lesson objectives

1.	identify the types of plane shapes and state their properties, similarities and differences
2.	identify the parts of a circle
3.	find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Plane shapes: types, properties and perimeter

Class: JSS1 **Subject:** Mathematics **Week:** 2

Section 1: Types of Plane Shapes and Their Properties

A **plane shape** — also called a **two-dimensional (2-D) shape** — is any flat shape that lies on a single surface. The plane shapes you will meet here are the triangle, the rectangle, the square, the parallelogram, the trapezium, the circle and the other polygons. In this section you will learn to recognise these shapes, state their properties, and see how they are similar and how they differ.

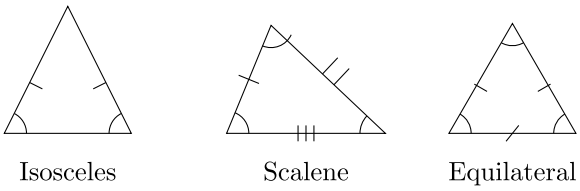
Triangles

A **triangle** is a closed shape with three straight sides and three corners (vertices). There are three special kinds of triangle, shown in fig-01. The small ticks on the sides mark the sides that are equal, and the arcs mark equal angles.

Isosceles	Scalene	Equilateral
(i) 2 sides are equal (the two slanted sides)	No side is equal to another	All 3 sides are equal
(ii) the two base angles are equal	No angle equals any other	All 3 angles are equal (60 ° each)
(iii) one line of symmetry	No line of symmetry	3 lines of symmetry

An **isosceles** triangle has two equal sides and two equal base angles, and it folds exactly along one line. A **scalene** triangle has no two sides equal, no two angles equal, and no line of symmetry. An **equilateral** triangle has all three sides equal and all three angles equal at 60 ° each, and it has three lines of symmetry.

Visual



Isosceles Scalene Equilateral

Isosceles: a single tick on each of the two slanted sides, an arc at each base angle. Scalene: one, two and three ticks on the three sides, an arc at each vertex. Equilateral: a single tick on each of the three sides, an arc at each vertex.

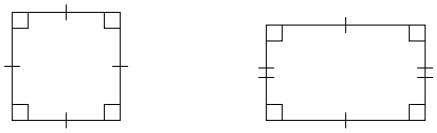
Squares and Rectangles

fig-02 shows a square and a rectangle. The small squares at the corners mark right angles (90 °), and the ticks mark the sides that are equal.

Squares	Rectangles
(i) All 4 sides are equal	Only the 2 opposite sides are equal
(ii) 2 pairs of parallel sides	2 pairs of parallel sides
(iii) All angles are 90 ° each	All angles are 90 ° each
(iv) 4 lines of symmetry	2 lines of symmetry
(v) Diagonals bisect it at a right angle	Diagonals bisect it but not at a right angle

A **square** has all four sides equal, all four angles 90 °, two pairs of parallel sides, and four lines of symmetry. Its diagonals cut each other at a right angle. A **rectangle** has only its opposite sides equal, all four angles 90 °, two pairs of parallel sides, and two lines of symmetry. Its diagonals bisect each other but not at a right angle. You can see the similarity: both are four-sided with two pairs of parallel sides and four right angles. The difference is that a square has all four sides equal while a rectangle has only its opposite sides equal.

Visual



Squares Rectangles

Square: a right-angle mark at each corner, a single tick on each of the four sides. Rectangle: a right-angle mark at each corner, a single tick on each of the two longer sides and a double tick on each of the two shorter sides.

Parallelograms and Trapezium

fig-03 shows a parallelogram and a trapezium. The arrows on the sides mark the sides that are parallel to each other.

Parallelogram	**Trapezium**
(i) 2 pairs of opposite sides are equal and parallel	Only one pair of sides are parallel
(ii) opposite angles are equal	opposite angles are not equal
(iii) no line of symmetry	no line of symmetry
(iv) diagonals divide it into equal triangles	diagonals divide it into two unequal triangles

A **parallelogram** has two pairs of opposite sides equal and parallel, and its opposite angles are equal. A **trapezium** has only one pair of parallel sides, so its opposite angles are not equal. Neither has a line of symmetry. The clearest difference between them is the number of pairs of parallel sides: a parallelogram has two, while a trapezium has only one.

Visual



Parallelogram Trapezium

Parallelogram: a single arrow on each of the two horizontal sides, a double arrow on each of the two slanted sides. Trapezium: a single arrow on each of the two parallel sides, and one diagonal drawn in.

Polygons

A **polygon** is a closed shape that has three or more straight sides. The triangles, squares, rectangles, parallelograms and trapezium you have met are all polygons. Polygons are named by their number of sides, and any polygon with n sides can be divided into $n-2$ triangles by drawing diagonals from one vertex.

Name of Polygon	**No. of sides**	**No. of triangles $(n-2)$**
Triangle	3	1
Quadrilateral	4	2
Pentagon	5	3
Hexagon	6	4
Heptagon	7	5
Octagon	8	6
Nonagon	9	7
Decagon	10	8
Hendecagon or undecagon	11	9
Dodecagon	12	10

So a triangle $(n=3)$ divides into $3-2=1$ triangle, a quadrilateral $(n=4)$ divides into $4-2=2$ triangles, and so on. This pattern lets you find the number of triangles for any polygon from the number of its sides.

Worked example

A polygon has 7 sides and all of its sides are equal.

1. What is its name? A 7-sided polygon is a **heptagon**.
2. How many triangles can it be divided into? Use $n-2$ with $n=7$: $7-2=5$ triangles.
3. If all its sides are equal, which triangle does it resemble? A polygon with all sides equal is like the **equilateral** triangle, which also has all sides equal.

Practice

1. A shape has all four sides equal, all four angles $90\,^\circ$, and two pairs of parallel sides. Which shape is it, and how many lines of symmetry does it have?
2. State one similarity and one difference between a parallelogram and a trapezium.
3. How many triangles can an octagon be divided into if you draw diagonals from one vertex to all non-adjacent vertices?
4. Name the triangle with one line of symmetry and two equal sides, and the triangle with no line of symmetry and no two sides equal.

Section 2: The Parts of a Circle

A circle is a special plane shape, and mathematicians have given names to its different parts. You can see them all on the two labelled circles in fig-04, so keep that figure in front of you as you read.

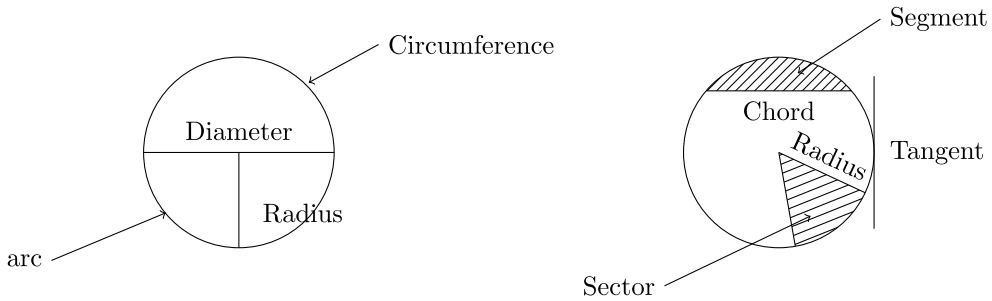
The **circumference** is the curved line that forms the boundary of the circle — it is the distance all the way round. Any curved part of the circumference is called an **arc**. Look at the first circle in fig-04: the arrow labelled "arc" points to a piece of the curved boundary.

The **centre** is the point in the very middle of the circle. A straight line that joins the centre to the circumference is called a **radius** (plural *radii*). In fig-04 you can see a radius drawn from the centre down to the circumference.

A straight line that joins any two points on the circumference is called a **chord**. A **diameter** is a special chord: it passes through the centre of the circle, so it is the longest chord and it cuts the circle exactly in half. In the first circle of fig-04, the line labelled "Diameter" passes through the centre and touches the circumference on both sides. Because it goes through the centre, a diameter is always exactly twice a radius.

There are three more parts shown on the second circle in fig-04. A **segment** is the part of the circle cut off by a chord — in the figure, the shaded region above the chord near the top is a segment. A **sector** is the part of the circle bounded by two radii and the arc between them — in the figure it is the shaded wedge, like a slice of a circle, with one of its sides labelled Radius. Finally, a **tangent** is a straight line that touches the circle at exactly one point; in fig-04 the line on the right labelled "Tangent" just touches the circle without cutting across it.

Visual



Two labelled circles. The first has the diameter drawn across it and a radius from its centre to the circumference, with the arc and the circumference labelled. The second has a chord near the top with the segment above it shaded, a shaded sector below with its right-hand bounding line labelled Radius, and a tangent touching it on the right.

Worked example — reading the parts from fig-04

On the first circle, the horizontal line drawn right across the circle through the centre is the **diameter**. The line from the centre straight down to the curved boundary is a **radius**. The curved part of the boundary is the **arc**, and the whole curved boundary is the **circumference**.

Now look at the second circle. The straight line near the top that joins two points on the circumference is a **chord**, and the shaded region above it is the **segment**. The shaded wedge below, bounded by two radii and the arc between them, is a **sector** — notice one of its straight sides is labelled Radius. The straight line on the right that touches the circle at one point only is the **tangent**.

Practice

1. a. Name the curved line that forms the whole boundary of a circle.
   b. What is the name of any straight line that joins the centre of a circle to its circumference?
   c. A straight line that joins two points on the circumference of a circle is called a _____. What special name do we give to a chord that passes through the centre?
   d. In fig-04, the shaded region above the chord near the top of the second circle is called a _____, and the shaded wedge below it is called a _____. What is the straight line on the right that touches the circle at exactly one point called?

Section 3: Finding the Perimeter of Plane Shapes

The perimeter of a shape is the distance all the way round it. To find it you simply add up the lengths of all the sides. Because some shapes have sides of equal length, you can often multiply instead of adding.

- A square has four equal sides, so its perimeter is $4 \times$ side.
- A rectangle has opposite sides equal (length L and breadth b), so its perimeter is $2L + 2b$, or $2(L + b)$.
- A parallelogram also has opposite sides equal, so its perimeter is $2a + 2b$ when the two different side lengths are a and b.
- A trapezium usually has four different sides, so you just add all four: $a + b + c + d$.
- A regular polygon has all sides equal, so its perimeter is (number of sides) $\times$ (length of one side).
- The circle has its own name for perimeter — the circumference — given by $C = 2\pi r$ or $C = \pi d$, where r is the radius and d is the diameter ($d = 2r$).

Let's work through each one.

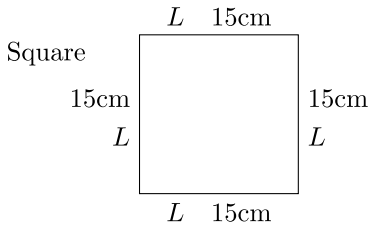
Example 1 — square. Calculate the perimeter of a square of side 15cm (see fig-05).

Every side is 15cm, and there are four of them:

$$P = L + L + L + L = 4L = 4 \times 15 = 60 \text{ cm}$$

So the perimeter is 60 cm.

Visual



A square, named *Square* at its left, with all four sides marked twice — L and 15cm.

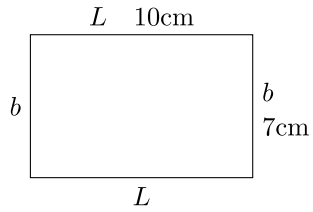
Example 2 — rectangle. Find the perimeter of a rectangle whose length is 10cm and breadth 7cm (see fig-06).

A rectangle has two lengths and two breadths:

$$P = L + b + L + b = 2L + 2b = 2(L + b)$$
$$= 2(10 + 7) = 2(17) = 34 \text{ cm}$$

So the perimeter is 34 cm.

Visual



A rectangle with both long sides marked L, the top one also 10cm, and both short sides marked b, the right-hand one also 7cm.

Example 3 — finding a missing side of a rectangle. The perimeter of a rectangle is 60cm and its length is 17cm. Calculate its breadth (see fig-07).

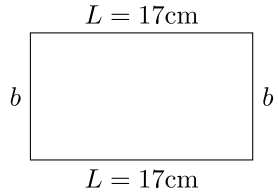
Write the perimeter formula and substitute what you know:

\[P = 2(L + b)$$ $$60 = 2(17 + b)\]
Divide both sides by 2:

$$30 = 17 + b$$
$$b = 30 - 17 = 13 \text{ cm}$$

So the breadth is 13 cm.

Visual



A rectangle with both long sides marked $L = 17$cm and both short sides marked b.

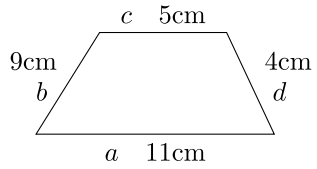
Example 4 — trapezium. Find the perimeter of the trapezium in fig-08, whose four sides are 11cm, 9cm, 5cm and 4cm.

Add all four sides:

$$P = a + b + c + d = 11 + 9 + 5 + 4 = 29 \text{ cm}$$

So the perimeter is 29 cm.

Visual



A trapezium with each side lettered and measured: the top c, 5cm; the left b, 9cm; the right d, 4cm; the base a, 11cm.

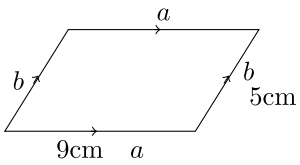
Example 5 — parallelogram. Find the perimeter of the parallelogram in fig-09, whose base a is 9cm and slanted side b is 5cm.

Opposite sides are equal, so:

$$P = a + a + b + b = 2a + 2b = 2(a + b)$$
$$= 2(9 + 5) = 2(14) = 28 \text{ cm}$$

So the perimeter is 28 cm.

Visual



A parallelogram with a single arrow on each of its four sides: the two horizontal ones both lettered a, the lower also measured 9cm, and the two slanted ones both lettered b, the right-hand one also measured 5cm.

Example 6 — regular hexagon. Find the perimeter of a regular hexagon whose sides are each 5.2cm (see fig-10).

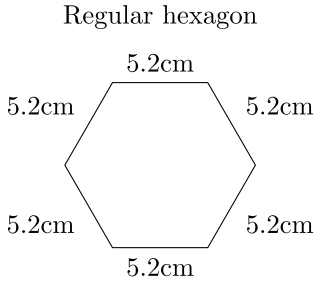
A hexagon has 6 equal sides, so:

$$P = 6 \times 5.2 = 31.2 \text{ cm}$$

So the perimeter is 31.2 cm.

The same idea works for any regular polygon: multiply the number of sides by the length of one side.

Visual



A regular hexagon, named above the drawing, with each of its six sides marked 5.2cm.

Example 7 — circle. Calculate the perimeter (circumference) of a circle of radius 7cm, using $\pi = 3\frac{1}{7}$ (see fig-11).

The perimeter of a circle is the circumference:

$$C = 2\pi r = 2 \times 3\tfrac{1}{7} \times 7$$

Write $3\frac{1}{7}$ as $\frac{22}{7}$:

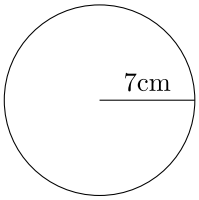
$$C = 2 \times \tfrac{22}{7} \times 7 = 2 \times 22 = 44 \text{ cm}$$

You get the same result with $C = \pi d$. Since $d = 2r = 2 \times 7 = 14$ cm:

$$C = \tfrac{22}{7} \times 14 = 22 \times 2 = 44 \text{ cm}$$

So the circumference is 44 cm.

Visual



A circle with a radius drawn from the centre to the right of the circumference, marked 7cm.

Example 8 — finding the radius from the circumference. The circumference of a circle is 440cm. Find its radius, using $\pi = \frac{22}{7}$ (see fig-13).

Use $C = 2\pi r$ and substitute $C = 440$:

$$440 = 2 \times \tfrac{22}{7} \times r = \tfrac{44r}{7}$$

Cross-multiply:

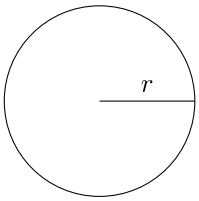
\[440 \times 7 = 44r$$ $$3080 = 44r\]

Divide both sides by 44:

$$r = \frac{3080}{44} = 70 \text{ cm}$$

So the radius is 70 cm.

Visual



A circle with a radius drawn from the centre to the right of the circumference, marked r.

Example 9 — a shape that is part straight, part curved. Calculate the perimeter of the shape in fig-14: two straight sides of 4.9cm joined by a quarter-circle arc of radius 4.9cm.

The curved part is one quarter of a full circle's circumference:

$$\text{curved part} = \frac{1}{4} \times 2\pi r = \frac{1}{4} \times 2 \times \frac{22}{7} \times 4.9$$

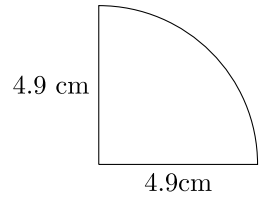
$$\frac{1}{4} \times \frac{44}{7} \times \frac{49}{10} = \frac{11}{7} \times \frac{49}{10} = \frac{11 \times 7}{10} = \frac{77}{10} = 7.7 \text{ cm}$$

Now add the two straight sides and the curved part:

$$P = 4.9 + 4.9 + 7.7 = 17.5 \text{ cm}$$

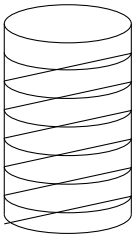
So the perimeter is 17.5 cm.

Visual



A quarter circle: an upright side and a base meeting at a right angle, each marked 4.9cm, with an arc closing them from the top of the upright to the end of the base.

Visual



An upright cylinder with a rope wound round it, drawn as a series of turns spiralling down the body from the top face to the base.

Practice

1. Find the perimeter of a square whose sides are 8cm.
2. Find the perimeter of a rectangle of length 7cm and breadth 4cm.
3. What is the perimeter of a regular pentagon whose sides are 5cm?
4. Find the perimeter of a parallelogram whose base is 6cm and slanted sides are 5cm.
5. The circumference of a circle is 22cm. Find the radius of the circle, using $\pi = \frac{22}{7}$.
6. A rectangle is such that its length is twice its breadth. If its breadth is d, find the total length of all its sides.
7. Find the perimeter of a regular hexagon whose sides are each 5cm.
8. Calculate the perimeter of the figure below: it has a base of 9cm, two straight sides of 5cm each, a top made of two straight pieces of 1cm each joined by a semicircular arc of radius 3.5cm (use $\pi = \frac{22}{7}$).

Check your work

Attempt every question before you read this page.

Types of Plane Shapes and Their Properties

1. A square; it has 4 lines of symmetry.

   The shape has all four sides equal and all four angles right angles, which matches the square. A square has four lines of symmetry.
2. Similarity: neither has a line of symmetry. Difference: a parallelogram has two pairs of parallel sides, while a trapezium has only one pair.

   Both shapes have no line of symmetry. The key difference is in parallel sides: a parallelogram has two pairs of equal and parallel opposite sides, while a trapezium has only one pair of parallel sides.
3. 6 triangles.

   An octagon has 8 sides, so $n = 8$. Using $n - 2$, the number of triangles is $8 - 2 = 6$.
4. The isosceles triangle has one line of symmetry and two equal sides; the scalene triangle has no line of symmetry and no two sides equal.

The Parts of a Circle

1. a. The circumference.
   b. A radius.
   c. A chord; a diameter.
   d. Segment; sector. The line is a tangent.
   e. The whole curved boundary of a circle is the circumference; any curved part of it is an arc.
   f. The radius joins the centre to the circumference. A diameter passes through the centre and is twice a radius.
   g. A chord joins any two points on the circumference. A chord that passes through the centre is a diameter.
   h. A segment is the part cut off by a chord; a sector is the part between two radii and their arc; a tangent touches the circle at exactly one point.

Finding the Perimeter of Plane Shapes

1. $4 \times 8 = 32$ cm

   A square has four equal sides, so $P = 4 \times 8 = 32$ cm.
2. $2(7 + 4) = 2(11) = 22$ cm

   $P = 2(L + b) = 2(7 + 4) = 2(11) = 22$ cm.
3. $5 \times 5 = 25$ cm

   A regular pentagon has 5 equal sides, so $P = 5 \times 5 = 25$ cm.
4. $2(6 + 5) = 2(11) = 22$ cm

   Opposite sides of a parallelogram are equal, so $P = 2a + 2b = 2(a + b) = 2(6 + 5) = 22$ cm.
5. 3.5 cm

   $C = 2\pi r$, so $22 = 2 \times \frac{22}{7} \times r = \frac{44r}{7}$. Cross-multiplying gives $44r = 22 \times 7 = 154$, so $r = 154 \div 44 = 3.5$ cm.
6. $6d$

   Length $L = 2d$ and breadth $b = d$. Perimeter $= 2L + 2b = 2(2d) + 2(d) = 4d + 2d = 6d$.
7. $6 \times 5 = 30$ cm

   A regular hexagon has 6 equal sides, so $P = 6 \times 5 = 30$ cm.
8. 32 cm

   The semicircular arc is half a circle of radius 3.5cm, so its length is $\pi r = \frac{22}{7} \times 3.5 = 11$ cm. Now add all the straight pieces: $9 + 5 + 5 + 1 + 1 = 21$ cm. Total perimeter $= 21 + 11 = 32$ cm.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-95432189	**Date**	
Subtopic	Plane shapes: types, properties and perimeter	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____